

Assignment – 9

Question – 1:

Code:

```
1/*
2Declare a pair of character arrays as char charArray1[10],
3charArray2[10]; Initialise the elements of charArray1 while
4declaring. Read the elements of charArray2 character by
5character from user. Print the elements of both the character
6arrays using a loop.
7*/
8
9#include <stdio.h>
10
11int main() {
12    char charArray1[10] = "AaryanKh07", charArray2[10];
13    printf("Enter 10 characters: ");
14    for(int i = 0; i < 10; i++) scanf("%c", &charArray2[i]);
15    printf("\nThe strings are: ");
16    for(int i = 0; i < 10; i++) printf("%c", charArray1[i]);
17    printf(" ");
18    for(int i = 0; i < 10; i++) printf("%c", charArray2[i]);
19    printf("\n");
20 }
```

Output:

```
students@cseLabIadnin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_9$ ./q1
Enter 10 characters: helloworld
The strings are: AaryanKh07 helloworld
```

Question – 2:

Code:

```
1/*
2Declare a character array as char charArray[10]; Assign the base
3address of the array to a character pointer. Read the elements of
4the character array and print the elements of the character array
5element by element using the pointer.
6*/
7
8#include <stdio.h>
9
10int main() {
11    char charArray[10], *p;
12    p = &charArray[0];
13    printf("Enter 10 characters: ");
14    for(int i = 0; i < 10; i++) scanf("%c", p+i);
15    printf("\nThe entered string is: ");
16    for(int i = 0; i < 10; i++) printf("%c", *(p+i));
17    printf("\n");
18 }
```

Output:

```
students@cseLabIadnin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_9$ ./q2
Enter 10 characters: helloworld
The entered string is: helloworld
```

Question – 3:

Code:

```
1 /*
2 Declare a character array as char charArray[10]. Read the array
3 using scanf() function and %s. Print the same using printf()
4 function. Observe what happens when scanf() and printf() is used
5 for reading and printing multi-word strings.
6 */
7
8 #include <stdio.h>
9
10 int main() {
11     char charArray[10];
12     printf("Enter 10 characters: ");
13     scanf("%s", &charArray);
14     printf("\nThe given string is: %s.", charArray);
15 }
```

Output:

```
students@cseLab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_9$ ./q3
Enter 10 characters: hello world

The given string is: hello.
students@cseLab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_9$ ./q3
Enter 10 characters: helloworld

The given string is: helloworld.
```

Question – 4:

Code:

```
1 /*
2 Declare a character array as char charArray[10]. Use the gets()
3 and puts() functions to read and print the string. Compare the
4 same with scanf() and printf() functions when multi-word strings
5 are read.
6 */
7
8 #include <stdio.h>
9
10 int main() {
11     char charArray[10];
12     printf("Enter 10 characters: ");
13     gets(charArray);
14     printf("\nThe given string is using gets() and puts() is: ");
15     puts(charArray);
16     printf("Enter 10 characters: ");
17     scanf("%s", &charArray);
18     printf("\nThe given string is using scanf() and printf() is: ");
19     printf("%s", charArray);
20     printf("\n");
21 }
```

Output:

```
students@cseLab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_9$ ./q4
Enter 10 characters: hello world

The given string is using gets() and puts() is: hello world
Enter 10 characters: hello worl

The given string is using scanf() and printf() is: hello
```

Question – 5:

Code:

```
1/*
2Read a string from the user and find length of the string using
3built in function as well as by counting the characters in the string.
4*/
5
6#include <stdio.h>
7#include <string.h>
8
9int main() {
10    char str[20];
11    printf("Enter a string: ");
12    gets(str);
13    int len1 = strlen(str), len2 = 0;
14    printf("\nThe length of given string using strlen() is %d.", len1);
15    for(int i = 0; str[i] != '\0'; i++) len2++;
16    printf("\nThe length of given string using loop is %d.", len2);
17}
18
```

Output:

```
students@cse1abiadnin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/Z4CSE1027/assignment_9$ ./q5
Enter a string: hello world

The length of given string using strlen() is 11.
The length of given string using loop is 11.
```

Question – 6:

Code:

```
1/*
2Read a string from the user and copy the string using built in
3function as well as by copying character by character.
4*/
5
6#include <stdio.h>
7#include <string.h>
8
9int main() {
10    char str[20], cstr1[20], cstr2[20];
11    printf("Enter a string: ");
12    gets(str);
13    int len = strlen(str);
14    strcpy(cstr1, str);
15    for(int i = 0; i < len; i++) cstr2[i] = str[i];
16    printf("\nThe copied strings using strcpy() is: %s", cstr1);
17    printf("\nThe copied strings by copying char by char is: %s\n", cstr2);
18}
```

Output:

```
students@cse1abiadnin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/Z4CSE1027/assignment_9$ ./q6
Enter a string: meow

The copied strings using strcpy() is: meow
The copied strings by copying char by char is: meow
```

Question – 7:

Code:

```
1 /*
2 Read 2 strings from user and compare the two strings, character
3 by character. Perform the same using built in function.
4 */
5
6 #include <stdio.h>
7 #include <string.h>
8
9 int main() {
10     char str1[20], str2[20];
11     printf("Enter a string: ");
12     gets(str1);
13     printf("Enter another string: ");
14     gets(str2);
15     int len1 = strlen(str1);
16     int len2 = strlen(str2);
17     int l = strcmp(str1, str2), is = 1;
18     if((len1 != len2) || l != 0) printf("The strings are not the same!\n");
19     for(int i = 0; i < len1; i++) {
20         if(str1[i] != str2[i]) is = 0;
21     }
22
23     if(is == 1) printf("The strings are the same!\n");
24     else printf("The strings are not the same!\n");
25 }
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_9$ ./q7
Enter a string: hello world
Enter another string: hello world
The strings are the same!
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_9$ ./q7
Enter a string: hello
Enter another string: jello
The strings are not the same!
```